Noise protocol used in stratum v2

Alice and Bob can have:



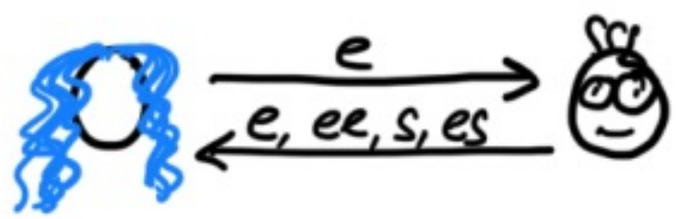
e = ephemeral key

s = static key

h = handshake hash (useful for auth)

ck = chaining key (for making transport msg ciphers)

NX handshake

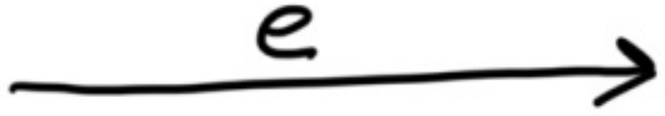


NX handshake

ALICE



initiator



BOB

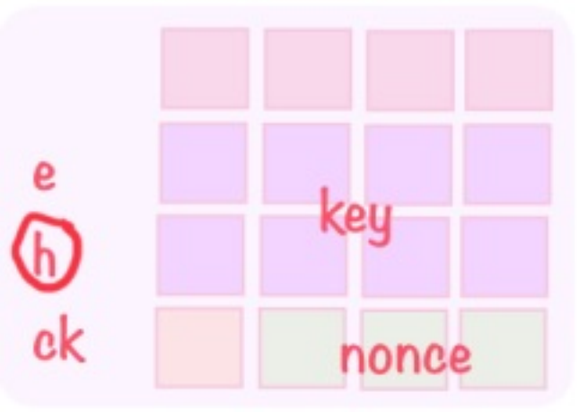


responder

Alice sends her e (unencrypted but encoded using ellswift as pseudorandom bytes)

operations:

- h is updated to include e by hashing



When Bob receives her e

operations:

- h is updated to include e by hashing



ALICE

INITIATOR

e, ee, s, es

BOB

RESPONDER

e

Bob sends his e (unencrypted but encoded using ellswift as pseudorandom bytes)

operations:
- h is updated to include e by hashing



ee

ecdh

alice's ephemeral public key

bob's ephemeral private key

HKDF is used to derive secure keys



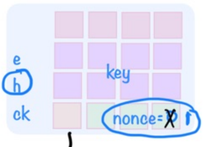
operations:
1. k is updated
2. ck is updated



send s

Bob sends his s (encrypted and encoded using ellswift as pseudorandom bytes)

operations:
- h up till now is used as aad in the encrypted static key message
- h is updated to include s by hashing



keystream to encrypt obtained from chacha20, nonce is incremented afterwards

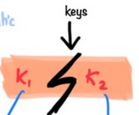
es

ecdh

alice's ephemeral public key

bob's static private key

HKDF is used to derive secure keys



operations:
1. k is updated
2. ck is updated



nonce is set to 0

CERTIFICATE

3rd party

CERTIFICATE consist of:

version	2 bytes
valid_from	4 bytes
valid_to	4 bytes
signature signed using pool authority key	64 bytes

Bob sends an encrypted certificate

operations:
- h up till now is used as aad in the encrypted certificate message
- h is updated to include the encrypted message by hashing

Finally, Bob uses the chaining key ck to derive keys to instantiate ciphers for sending and receiving transport messages

